



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2013

BIOLOGY

9648/01

HIGHER 2

20 September 2013

Paper 1 Multiple Choice

1 hour 15 minutes

Additional Material: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, pencil clips, highlighters, glue or correction fluid.

Write your name, Centre number and index number on the Answer Sheet provided.

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

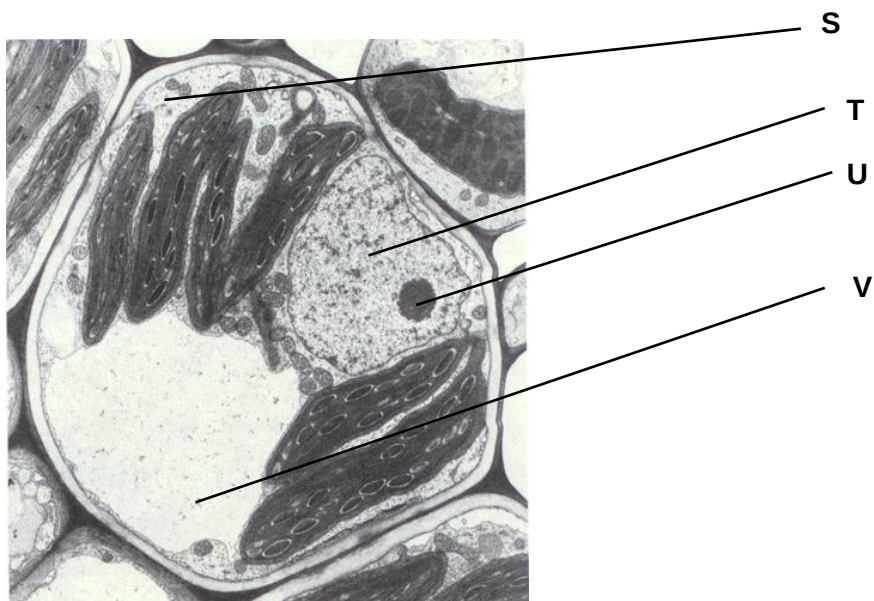
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

Calculators may be used.

This question paper consists of **23** printed pages.

- 1 If the cell shown below has been cultured in an environment where the source of nitrogen provided is labelled radioactively, in which of the following labelled regions would we expect to find high radioactivity?



- A S and T only
 B T and U only
 C S, T and U only
 D S, T, U and V
- 2 Which row of molecules found in the cell surface membrane have their corresponding roles listed?

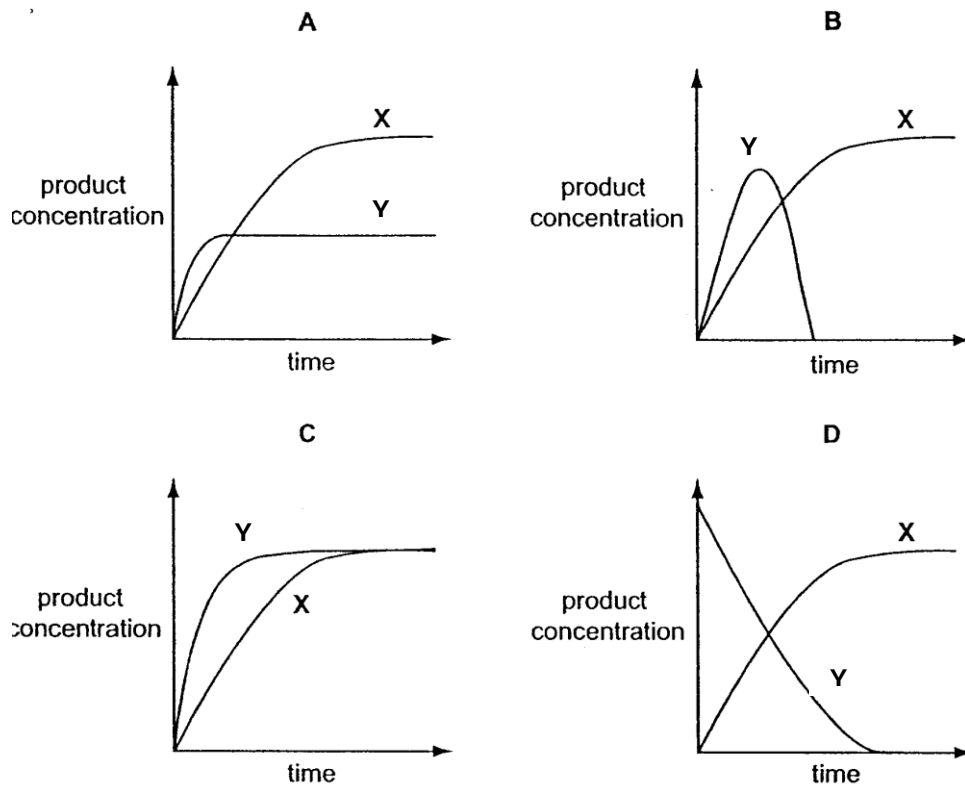
	Act as receptor sites for hormones	Form hydrogen bonds with water	Allow passage of ions	Cell to cell adhesion and recognition
A	Glycoproteins and glycolipids	Phospholipids and glycoproteins	Glycoproteins and glycolipids	Glycoproteins and glycolipids
B	Glycoproteins and proteins	Phospholipids and glycoproteins	Glycoproteins and proteins	Glycoproteins and glycolipids
C	Glycoproteins and phospholipids	Glycoproteins and glycolipids	Glycoproteins and glycolipids	Glycoproteins and cholesterol
D	Glycoproteins and glycolipids	Glycoproteins and cholesterol	Glycoproteins and phospholipids	Glycoproteins and cholesterol

- 3 During the production of fruit juice, enzymes are used to break down the components of cell walls. Which carbohydrate will be produced by this hydrolysis?

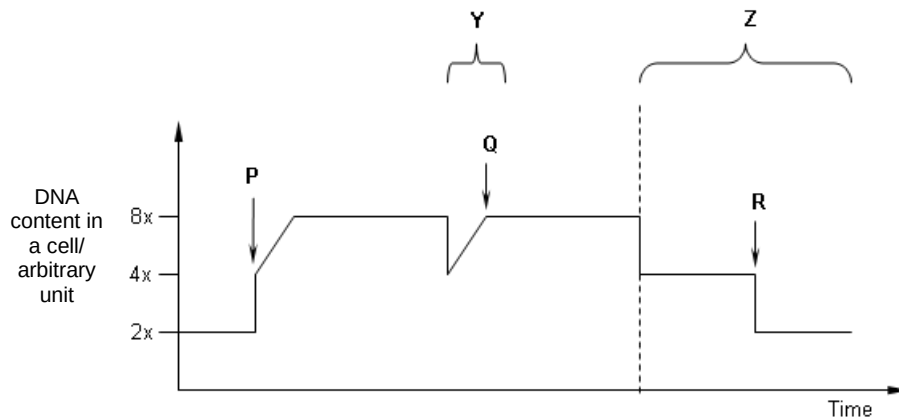
A Sucrose
 B Maltose
 C α - glucose
 D β - glucose

- 4 Two enzyme experiments were carried out. The first, experiment X, was carried out at a constant temperature of 37°C. During the second experiment, Y, the temperature was increased from 37°C to 80°C.

Which graph shows the results?



- 5 A cell in the G1 phase has two homologous pairs of chromosomes. It then undergoes a mitotic division, followed by meiosis. At the end of meiosis II, what is the total number of chromosomes and gene loci found in all the daughter cells formed?
- A 8 chromosomes and 4 times as many gene loci as the original parent cell
 B 8 chromosomes and 8 times as many gene loci as the original parent cell
 C 16 chromosomes and 4 times as many gene loci as the original parent cell
 D 16 chromosomes and 8 times as many gene loci as the original parent cell
- 6 The graph represents the changes in the DNA content within a cell at different stages in the cell cycle.

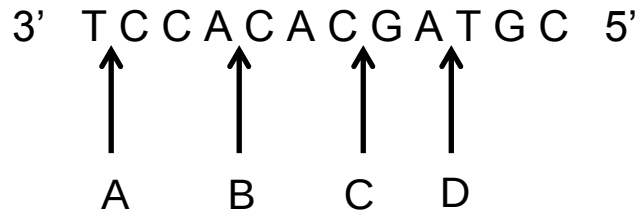


Name the events occurring at **P**, **Q** and **R**, and identify the stage where meiosis is occurring.

	P	Q	R	Meiosis occurring at
A	S phase	Fertilisation	Cytokinesis	Y
B	Fertilisation	Interphase	Cytokinesis	Z
C	S phase	Prophase	Telophase	Y
D	Fertilisation	Metaphase	Telophase	Z

- 7 The RNA triplet UAG acts as a stop codon terminating the synthesis of a polypeptide. The diagram shows a template strand of DNA which codes for four amino acids.

Where would a mutation, introducing a thymine nucleotide, result in the termination of translation?



- 8 Part of the amino acid sequences in normal and sickle cell haemoglobin are shown.

Normal haemoglobin
-thr-pro- glu-glu

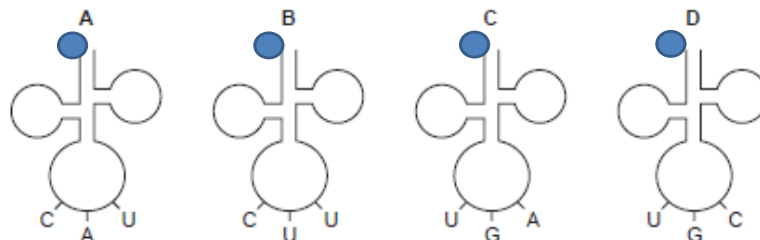
sickle cell haemoglobin
-thr-pro-val-glu

Possible mRNA codons for these amino acids are

glutamine (glu) GAA GAG
threonine (thr) ACU ACC

proline (pro) CCU CCC
valine (val) GUA GUG

Which tRNA molecule is **not** involved in the formation of this part of the sickle cell haemoglobin?



- 9 Haemoglobin is a globular protein consisting of four polypeptide chains – two alpha chains and two beta chains. In normal individuals, the gene for beta chain codes for glutamic acid at the sixth triplet.

In individuals with sickle cell anaemia, this base triplet is changed and codes for valine.

Which aspect of the haemoglobin molecule does this mutation directly affect?

- A the iron content
- B the primary structure
- C the quaternary structure
- D the secondary structure

10 Which of the following statements about the HIV is **not** true?

- A** The viral nucleotides contain ribose.
- B** HIV contains uracil, not thymine.
- C** gp 120 and gp 41 enable HIV to enter host by fusion.
- D** HIV makes the host cell produce reverse transcriptase.

11 How do viruses cause disease in animals?

- I** They inhibit normal host cell DNA, RNA or protein synthesis.
- II** They disrupt and inactivate the tumor suppressor genes of the host cell causing uncontrolled cell division.
- III** They disrupt and inactivate the oncogenes of the host cell causing uncontrolled cell division.
- IV** Their viral proteins and glycoproteins on the surface membrane of host cells cause them to be recognized and destroyed by the body's immune system.
- V** They deplete the host cell of cellular materials essential for metabolic functions.

- A** I, II and V
- B** I, II, III and V
- C** I, II, IV and V
- D** All of the above

12 A bacterial cell can transfer DNA during conjugation when it

- A** contains an F factor.
- B** has been infected by a bacteriophage.
- C** is virulent.
- D** integrates into host chromosome.

- 13** When the *lac* operon for lactose metabolism is switched off, which of the following genes would still be expressed?
- I β -galactosidase gene
 - II RNA polymerase gene
 - III CAP gene
 - IV Repressor gene
- A** I and II
 - B** I and III
 - C** II, III and IV
 - D** All of the above
- 14** Which of the following statements correctly describes the control of transcription for genes in a *lac* operon?
- A** The genes are transcribed as a single transcription unit, with each gene having its own promoter.
 - B** Methylation of DNA causes the DNA to be tightly packed, hence RNA polymerase will not have access to the promoter region.
 - C** The genes have control elements such as enhancers that enable activators to bind, hence affecting the rate of transcription.
 - D** The genes are found next to one another on the same chromosome, and repressor proteins act as a form of transcriptional control.
- 15** Housekeeping genes are expressed all the time in all cells, whereas other non-essential genes are expressed only when needed. Under certain conditions, gene expression of selected genes can be increased.

Which of the following will increase gene expression in a eukaryotic cell?

- A** A chemical is added which will increase the activity of enzyme poly-A polymerase.
- B** Alternative splicing links different combinations of exons together in mRNA.
- C** Histone methylation results in the formation of heterochromatin.
- D** Activator proteins will bind to silencers to speed up the rate of transcription.

- 16** Gene expression in eukaryotes can be regulated at the translational level.

Which combination of statements correctly describes eukaryotic translational control?

	Condition	Effect
A	Lack of translation initiation factor proteins	Inhibition of translation of selected mRNA
B	Presence of repressor proteins binding to 5'-UTR of selected mRNA	Prevents small ribosomal subunit from binding
C	Presence of repressor proteins to distal control elements	mRNA is translationally-repressed
D	mRNA with a long poly-A tail	mRNA is degraded slowly by restriction endonuclease

- 17** A human papillomavirus (HPV) infection may cause cervical cancer in women. Which of the following is **not** a reasonable explanation for cancer development caused by HPV infection?

- A** HPV integrates its viral genes into host cell's genome.
- B** HPV may carry oncogenes that promote cell division.
- C** HPV may induce apoptosis of host cells.
- D** HPV promotes ubiquitination of p53 protein.

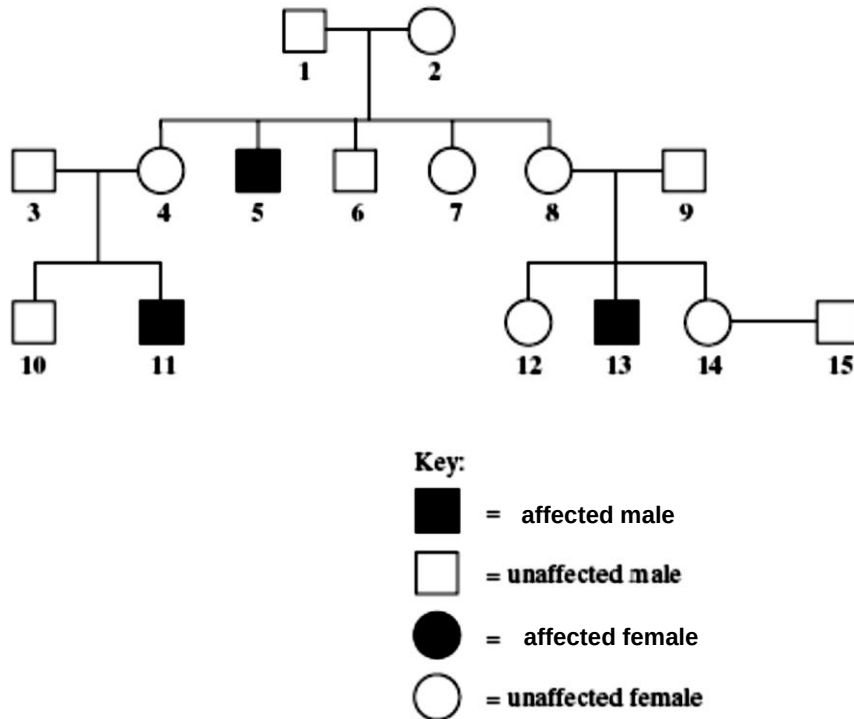
- 18** After explaining about cell structure, DNA, chromosomes, telomeres and centromeres in great detail during lectures and tutorials, a teacher decided to conduct a simple test to determine the level of understanding in students. The following statements were observed in the students' scripts.

- I Telomeres shorten after every cell division as a result of the end-replication problem in eukaryotic cells.
- II Telomeres are only found in eukaryotic chromosomes, since the prokaryotic chromosome is circular and not linear.
- III Centromeres are the protein molecules found at right angles to each other in centrosomes, and they are absent in plant cells.
- IV Telomerase is the enzyme that degrades telomeres, hence telomerase gene is switched off in adult somatic cells.

Which of the students' statements are correct?

- A** I and II only
- B** III and IV only
- C** I, II and IV only
- D** All of the above

- 19 A genetic disorder causes degeneration of muscle tissue in humans. The diagram shows the inheritance of such a genetic disorder in one family.



Which of the following statements can be concluded based on the diagram above?

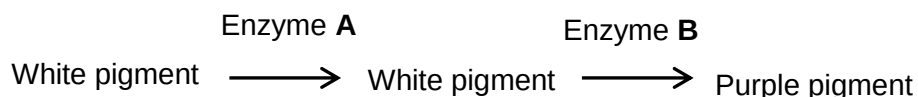
- A Individual 5 was affected with the genetic disorder as a result of genetic mutation during his lifetime.
- B Individual 10 should be an affected male too, since his brother (individual 11) was affected with the genetic disorder.
- C This genetic disorder is a sex-linked recessive trait.
- D This genetic disorder is due to an autosomal dominant allele.

- 20 A plant researcher tried to investigate a cross between two heterozygous Snapdragon plants that produced red flowers. She predicted three possible phenotypic outcomes, namely plants with white flowers, pink flowers and red flowers, with a phenotypic ratio of 4:3:9 respectively. When the cross was performed, she found 50 plants with white flowers only, 41 plants with pink flowers, and 85 plants with red flowers. A chi-squared test was performed, and the chi-squared value was calculated to be 4.74.

Degree of freedom	Probability, P				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

Which of the following statements is correct?

- A The degree of freedom is 3.
- B The calculated chi-squared value is greater than the critical chi-squared value.
- C There is a high probability that the difference between the observed and expected values is due to chance.
- D The probability that the difference between observed and expected values is due to chance is less than 5%.
- 21 A PhD student performed a cross between a pure-breeding plant that bore white flowers, and another pure-breeding plant that also bore white flowers. All the F_1 plants obtained bore purple flowers. Upon selfing the F_1 plants, a F_2 phenotypic ratio of 9 purple flowers: 7 white flowers was obtained. Gene **A** codes for the production of enzyme **A**, and gene **B** codes for enzyme **B**. The diagram below shows the metabolic pathway involved.



Which of the following statements **cannot** be deduced from the above information?

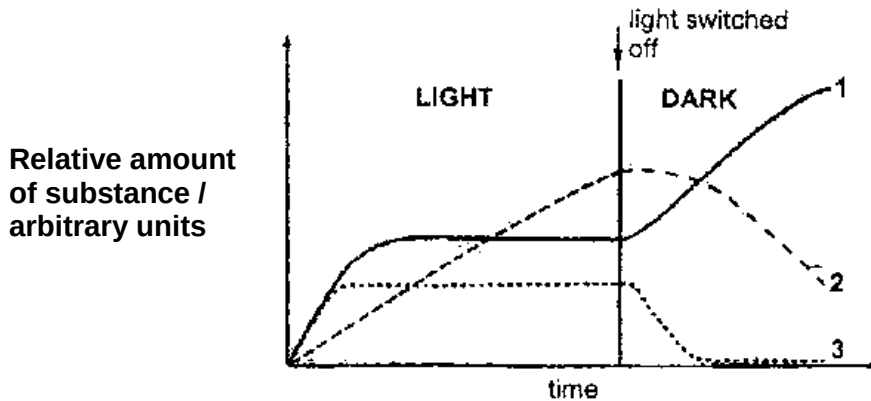
- A The parental pure-breeding plants must have the genotype AAbb and aaBB.
- B Gene **A** is epistatic to gene **B**.
- C Genes **A** and **B** are found on the same chromosome.
- D All the F_1 plants have the genotype AaBb.

- 22** A genetic cross was performed between two heterozygous *Drosophila* fruit flies with normal wings and red eyes. The allele for normal wings is dominant to the allele for vestigial wings, and the allele for red eyes is dominant to the allele for white eyes.

What is the probability of obtaining offspring that are pure-breeding?

- A** 1 out of 16
 - B** 3 out of 16
 - C** 4 out of 16
 - D** 9 out of 16
- 23** Which of the following statement(s) is/are correct?
- I** Polygenic inheritance is when a single phenotype is controlled by two or more sets of alleles.
 - II** The effect of environment has minimal influence in the expression of phenotype in polygenic inheritance.
 - III** Polygenic inheritance results in distribution of phenotypes that follow a normal distribution curve.
- A** I only
 - B** I and II only
 - C** I and III only
 - D** All of the above
- 24** Which of the following statements correctly compares oxidative phosphorylation and non-cyclic photophosphorylation?
- A** Both types of phosphorylation produce ATP and oxygen as end products.
 - B** Both types of phosphorylation produce ATP and the reduced form of a redox reagent.
 - C** Oxidative phosphorylation is involved in the conversion of one form of chemical energy to another while non-cyclic photophosphorylation is involved in converting light energy to chemical energy.
 - D** Water is an electron donor in non-cyclic photophosphorylation while it is an electron acceptor in oxidative phosphorylation.

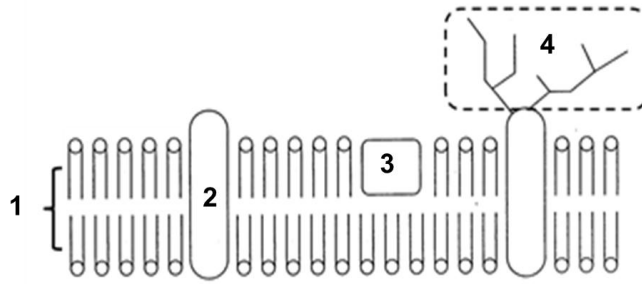
- 25 ^{14}C -labelled carbon dioxide was supplied to photosynthesising algae. The relative amounts of three organic compounds were measured. The diagram shows the results.



Which of the following are correct explanations for the graph above?

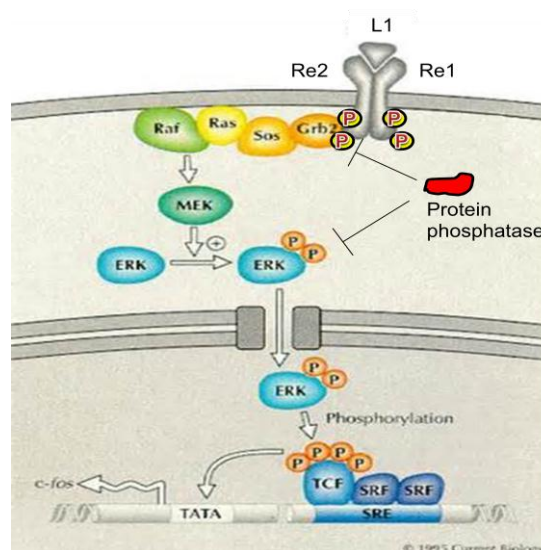
- I GP level falls as shown in graph 2 due to the absence of reduced NADP when light is switched off.
 - II GP level rises as shown in graph 1 due to the absence of ATP when light is switched off.
 - III Levels of RuBP and GP are constant during periods of light as they serve as intermediates in the Calvin cycle.
 - IV RuBP level falls as carboxylation of RuBP is independent of light as shown in graph 3.
 - V Sucrose level falls as shown in graph 3 due to the absence of ATP and reduced NADP.
- A** I, II and V only
B I, II and III only
C II, III and IV only
D III, IV and V only

- 26 The diagram below represents a cell surface membrane.



Which biomolecules shown above are amphipathic?

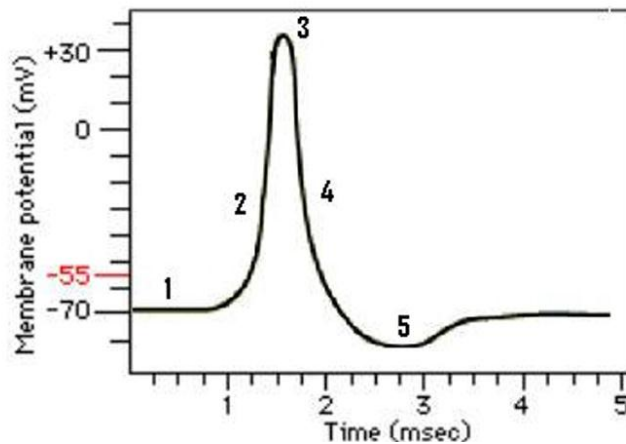
- A 1, 2 and 4 only
 - B 1, 2 and 3 only
 - C 2 and 3 only
 - D 3 and 4 only
- 27 The diagram below shows the cell signalling pathway involving a growth factor receptor.



From the given diagram, which step is involved in the role of signal amplification?

- A Binding of L1 to Re1 and Re2
- B Cross phosphorylation of Re1 and Re2
- C Phosphorylation of ERK by MEK
- D Dephosphorylation by protein phosphatase

28 The diagram below shows the events of an action potential.



Which of the following statements are correct?

- I. Events 1 to 5 occur in the synaptic knob during the transmission of an action potential across the synapse.
- II. Event 3, 4 and 5 are critical to ensure that the action potential travels unidirectionally across a synapse.
- III. Event 5 prevents a weak signal from initiating an action potential.
- IV. Event 1 is necessary for an impulse to be transmitted along an axon.

- A I and II only
- B I and III only
- C II and IV only
- D III and IV only

29 Which of the following statements does **not** correctly compare the neutral theory of molecular evolution and natural selection?

- A Neutral theory of molecular evolution accounts for most of the differences that we observe at the phenotypic level as compared to natural selection.
- B Neutral theory of molecular evolution accounts for most of the differences that we observe at the genotypic level as compared to natural selection.
- C The rate of change in the nucleotide sequence brought about by neutral theory of molecular evolution occurs at a constant rate due to random chance events while the rate of change brought about by natural selection can be fast or slow depending on the strength of the selection pressure.
- D Neutral theory of molecular evolution is largely responsible for the RFLP that we observe between species as compared to natural selection.

- 30 The figure below shows the alignment of amino acid residues of the alpha haemoglobin chain of the cow and the sheep. There are a total of 142 amino acid residues in both the alpha chain of the cow and sheep. Regions of polypeptide that form an alpha helix are underlined while positions where amino acid residues differ between the cow and the sheep alpha haemoglobin chain are marked with an arrow.

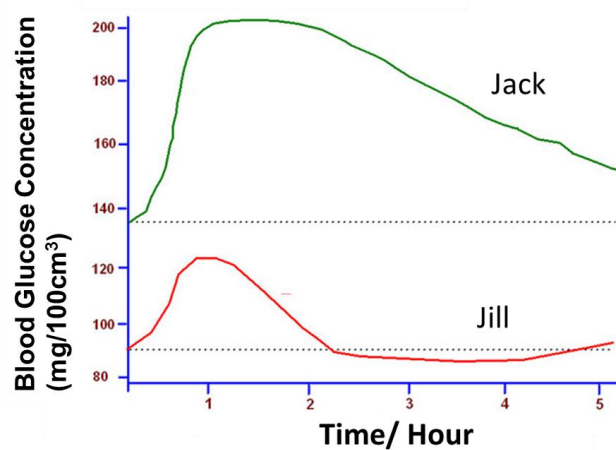
Amino acid position		Amino acid position
1	MVLSAADKGNVKAAWGKVGGHAAEYGAELERMFLSFPTTKTYFPHFDLSHGSAQVKGHG	60 Cow
1	MVLSAADKSNVKAAWGKVGGNAGAYGAELERMFLSFPTTKTYFPHFDLSHGSAQVKGHG	60 Sheep
61	AKVAAALTKAVEHLDLPGALSELSDLHAHKLRVDPVNFKLLSHSLLVTLASHLP	120 Cow
61	EKVAAALTKAVGHLDLPGTLSDLSDLHAHKLRVDPVNFKLLSHSLLVTLACHLP	120 Sheep
121	AVHASLDKFLANVSTVLTSKYR	142 Cow
121	AVHASLDKFLANVSTVLTSKYR	142 Sheep

Which of the following statements can be concluded from the diagram shown above?

- I. Differences between the amino acid residues between the 2 polypeptides are due to neutral mutation brought about by the degeneracy of the genetic code.
- II. Differences between the amino acid residues between the 2 polypeptides are due to neutral mutation brought about by mutations in the introns.
- III. Natural selection is one of the main driving forces that bring about the differences between the 2 polypeptides.
- IV. Differences between the amino acid sequences did not change the secondary structure of protein.

- A I and II only
- B I and III only
- C II and IV only
- D III and IV only

- 31 The diagram below shows the blood glucose concentration of two persons after eating a meal.

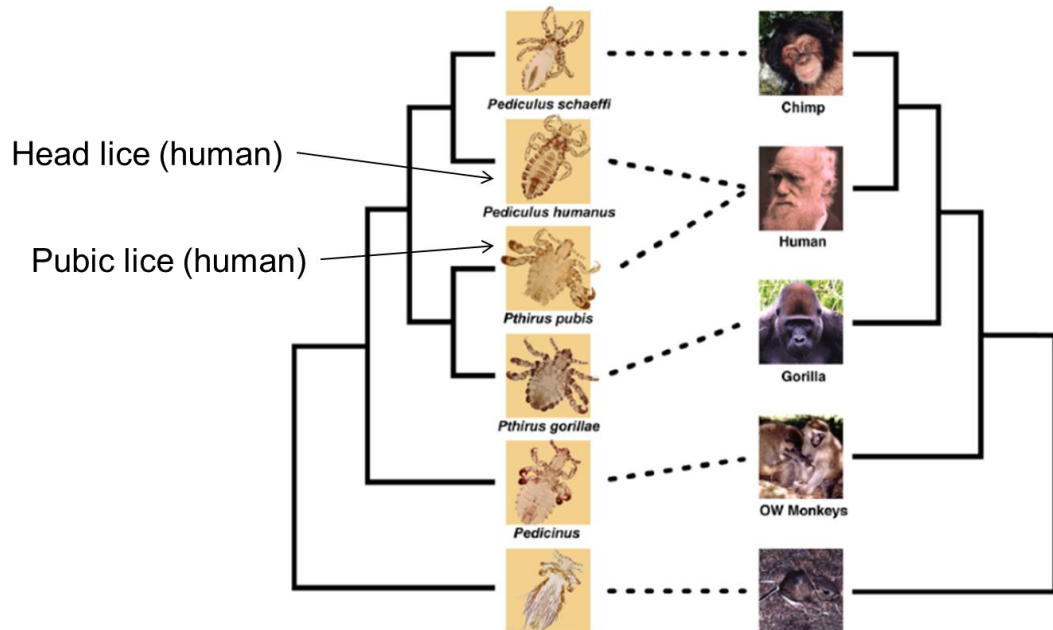


How can the levels of blood glucose concentrations in Jack and Jill be best explained?

- I. Jack and Jill both had insulin released into their bloodstream by their pancreas after the meal.
- II. The number of glucose transporter in the cell surface membrane of the muscle cells increased in Jill but not in Jack.
- III. Lowering of blood glucose level in Jack is brought about by utilisation of glucose in respiration and excretion through urine.
- IV. Jack must have type II diabetes while Jill is a healthy individual.

- A I and II only
- B II and III only
- C I and IV only
- D II, III and IV only

- 32 The figure below shows the phylogeny of selected primates and their lice which live on them. The dotted lines indicate the host of the respective lice.



Which of the following statement(s) can be concluded from diagram above?

- I. The common ancestor of *Pediculus humanus* and *Pediculus schaeffi* underwent allopatric speciation after the divergence of chimpanzee and humans.
- II. *Pthirus pubis* and *Pediculus humanus* underwent sympatric speciation on the human host.
- III. *Pthirus pubis* and *Pediculus humanus* are reproductively isolated from each other due to their ecological niches.
- IV. Gorillas and humans once stayed in close physical proximity, allowing pubic lice to change host.

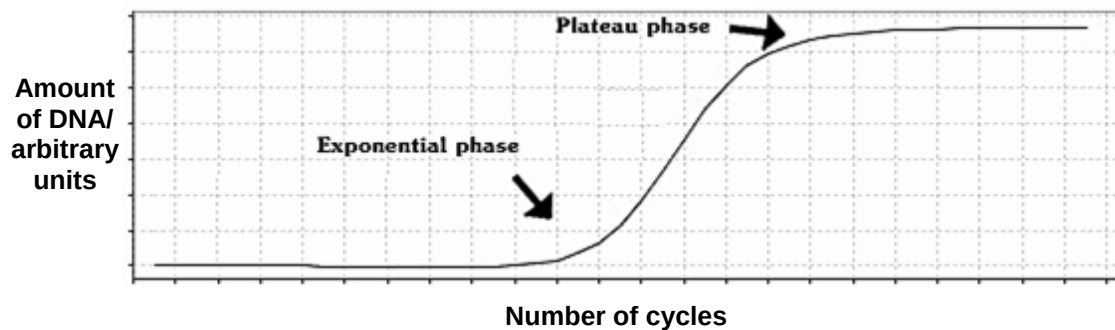
- A I only
- B I and II only
- C II and III only
- D I, III and IV only

- 33 The figure below shows how a metalmark moth mimics a jumping spider. It does so by having hairs on both its hind wings and forewings.



Which of the following statements **best** explains the evolution of spider mimicry in metalmark moth?

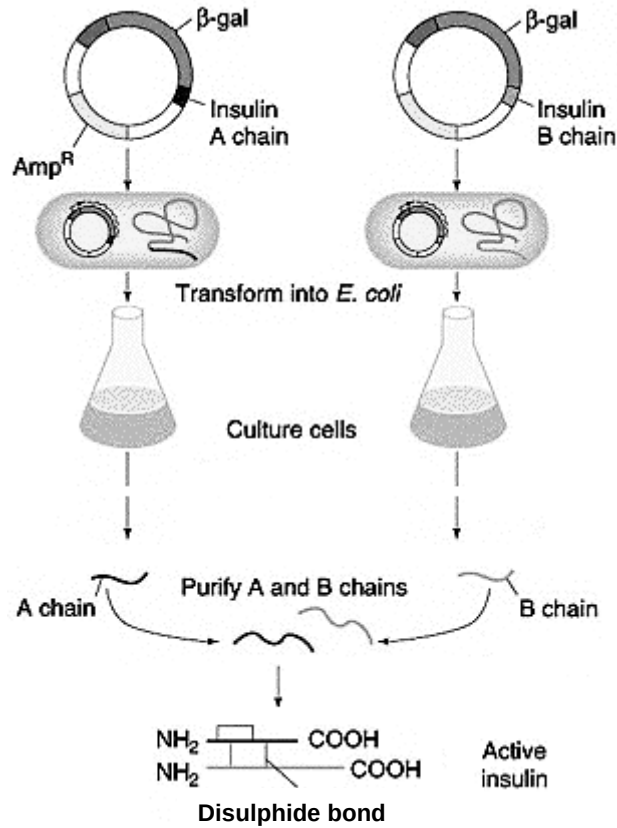
- A Moths mutate their genes to grow hairs on their wings in order to escape predation.
 - B Moths develop hairs on their wings to mimic spiders.
 - C Moths with hairs on their wings leave behind a higher proportion of offspring than moths without hairy wings.
 - D Moths with hairy wings experience a higher selection pressure than moth without hairy wings.
- 34 During the process of polymerase chain reaction (PCR), the amount of DNA synthesised can be traced using fluorescent probes and the measurements are shown in the following plot. The process initially goes through an exponential phase, followed by a plateau phase eventually.



Which of the following statements is **true**?

- A During the exponential phase, the number of DNA molecules synthesized after 15 cycles is 15^2 .
- B During the exponential phase, the temperature is always maintained at the optimum temperature of 72°C hence there is rapid amplification.
- C During the plateau phase, the reaction mixture is being depleted of ribonucleotides.
- D During the plateau phase, *Taq* polymerase may be denatured.

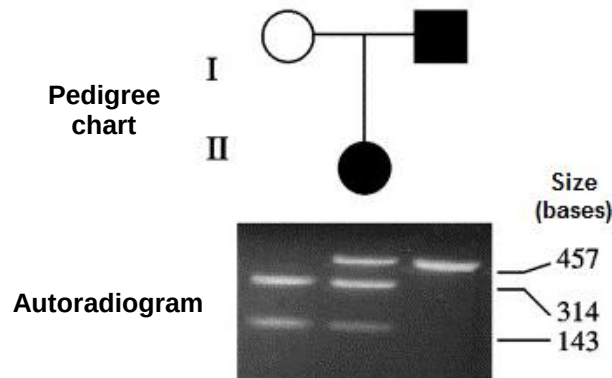
- 35** The insulin hormone is made up of two different polypeptide chains, A-chain and B-chain, joined together by disulphide bonds. The human insulin gene codes for both polypeptide chains. To produce the insulin hormone using *E. coli*, cDNA coding for each polypeptide chain are inserted into separate plasmids and used to transform separate batches of bacteria. The synthesised polypeptide chains are then mixed together to form the hormone.



Which one of the following reasons explains why the two chains have to be produced separately?

- A** Bacterial RNA polymerases cannot recognise eukaryotic promoters.
- B** Bacterial cells are not able to carry out post-translational modification of insulin.
- C** Bacterial cells are unable to excise introns from eukaryotic RNA.
- D** Bacterial ribosomes can only translate polycistronic mRNA.

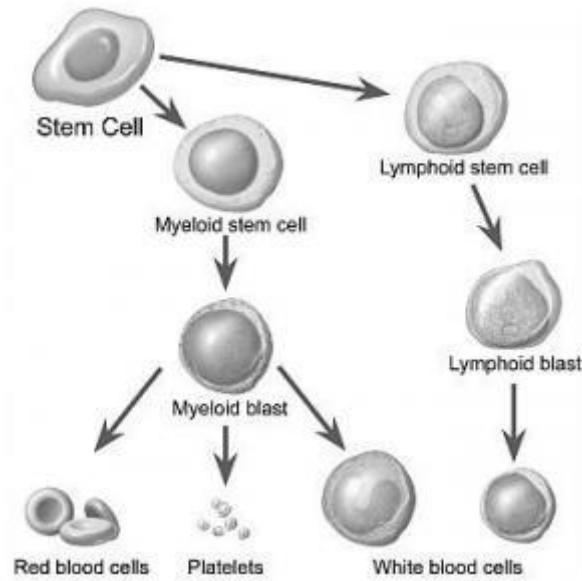
- 36** A family with a history of a genetic disease is studied using restriction fragment length polymorphism (RFLP) analysis. The mutation responsible for the disease allele produces a RFLP which can be detected by Southern blotting using an appropriate probe. The pedigree chart of the family is aligned with the autoradiogram obtained from Southern blotting. (Shaded symbols in the pedigree chart indicate individuals affected by disease.)



Based on the information given, which of the following can be deduced?

- A** The disease allele is dominant to the normal allele.
 - B** The mutation creates a new restriction site in the affected gene.
 - C** One of the parents in generation I is a carrier.
 - D** The offspring in generation II is a carrier.
- 37** The Human Genome Project (HGP) has brought about great advancements in health and medicine. Which of the following applications is **not** a result of the knowledge gained from the HGP?
- A** Designing of new antibody-based medicines which target proteins coded for by oncogenes.
 - B** Determining whether an individual is a suitable candidate for working in a nuclear power plant due to a genetic predisposition to cancer.
 - C** Predicting whether an individual may suffer adverse side effects from taking a particular medicine due to the individual's inability to break down the medicine.
 - D** Testing the gender of a fetus by detecting the presence or absence of the fetal Y chromosome in maternal blood.

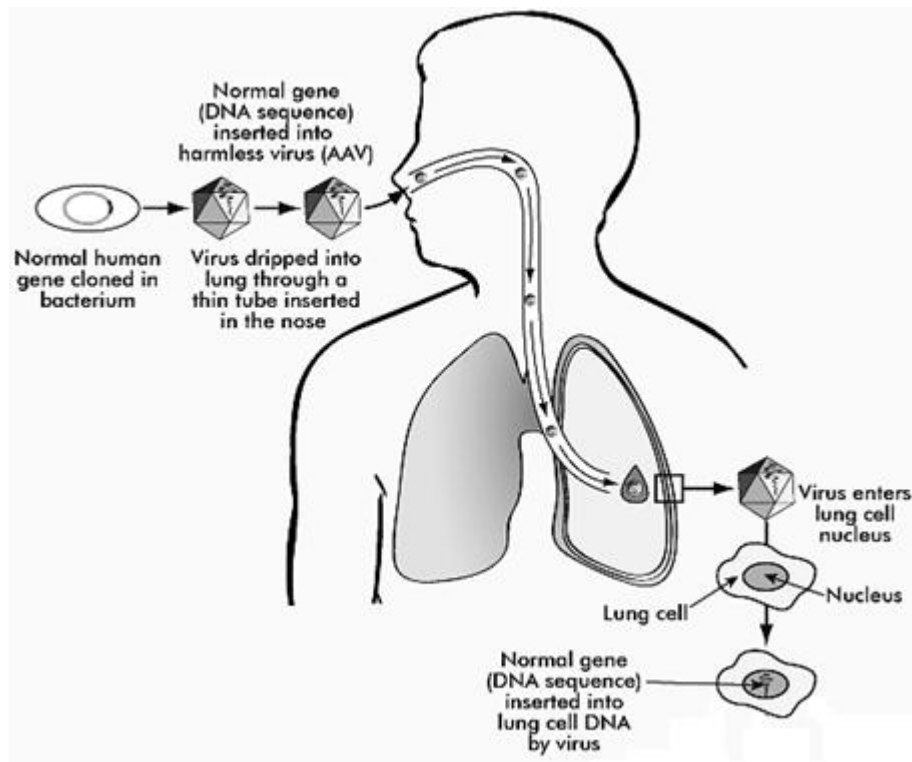
- 38 The following diagram shows how a stem cell can differentiate into different specialized cell types.



Which of these statements is **false** with regards to the stem cells shown?

- A The stem cells are multipotent.
 - B The stem cells can be found in both a fetus and an adult body.
 - C The stem cells can differentiate into the three germ layers in the adult body.
 - D The stem cells may be used in a bone marrow transplant to treat a patient with leukemia, a form of blood cancer.
- 39 Cholera is a disease caused by the bacteria *Vibrio cholerae* and is characterised by incessant diarrhoea and vomiting. Scientists have discovered a way to genetically modify rice plants such that these plants can produce part of the cholera toxin, which acts as a vaccine when consumed. Which statement is **not** a valid social or ethical implication of such edible vaccines?
- A The mass production of edible vaccines by rice plants lowers production costs compared to traditional pharmaceutical vaccines.
 - B The genes coding for the cholera toxin may spread to wild plant species through cross-pollination, causing the development of "superweeds".
 - C Antibiotic resistance marker genes used in the process of cloning the cholera toxin gene may lead to the development of antibiotic resistance in gut bacteria.
 - D The insertion of a foreign gene into the rice plant genome may lead to the production of unknown toxic components.

- 40 The following diagram shows one method where gene therapy can be used to treat cystic fibrosis. Inactivated adeno-associated virus (AAV) is used to introduce the normal gene into lung epithelial cells.



Which statement correctly describes the procedure shown?

- A An *ex vivo* mode of therapy is used.
- B Effects of the treatment by the inserted gene will not be permanent.
- C Somatic stem cells are specifically targeted by the virus.
- D Introduction of the normal gene into the host cell is less efficient using this method compared to using a liposomal vector.

End of Paper